Evaluation-Driven State Revision and the Encoding Hierarchy Under Bounded Context

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Abstract

We prove two linked results for any entity persisting under bounded active context with survival pressure. First, agency ($\gamma > 0$) necessarily produces evaluation-driven state revision, gradient descent on uncertainty U, with convergence guaranteed by a scalar Lyapunov argument: $U \geq 0$ is bounded below, each update reduces $\mathbb{E}[U]$ because novel encounters produce informative feedback, and monotone convergence applies without dimensional decomposition. The gradient never terminates under positive novelty rate: as long as the entity encounters states where $U > 0$, actions with positive information gain exist. Second, the certainty threshold for encoding an action at level L_i is $\theta_i = 1 - \varepsilon/C_i$, where C_i is the cost of failure at that level. This derives the learning rate hierarchy: evolutionary rates are necessarily orders of magnitude lower than individual learning rates, not by assumption but from cost-of-failure differences.

1 Introduction

An entity with agency $\gamma > 0$ acts on the world and receives feedback. What is the necessary structure of learning? Papers 1–3 in this series established that any entity persisting under bounded active context must maintain a directed graph over propositions [10], that control architectures converge to escalation at scale [11], and that knowledge and action are inseparable under survival pressure [12]. What remains open is two questions: what *dynamics* govern the updating process, and what determines the *threshold* at which an action earns promotion to a more permanent encoding level.

This paper answers both. We prove that the updating process is gradient descent on uncertainty U, with convergence guaranteed by a scalar Lyapunov argument that requires no dimensional decomposition (Theorem 1). We then prove that the certainty threshold for encoding at level L_i is determined by the cost of failure at that level, which derives the learning rate hierarchy across encoding levels (Theorem 3).

Several lines of prior work converge on these questions from different angles. Dupoux, LeCun, and Malik [3] propose a bilevel optimization framework (evolution as outer loop, development as inner loop) but treat the learning rate relationship between levels as a design parameter rather than deriving it. Cover and Thomas [2] provide the information-theoretic foundations that anchor our convergence argument. Schmidhuber [15] identifies curiosity as compression progress, structurally identical to our learning rate η_M. Hutter [8] constructs AIXI as an idealized agent but assumes unbounded computation, precisely the constraint our framework takes as central. Chollet [1] defines intelligence as skill-acquisition efficiency, close to η_M but treating prior knowledge as confound rather than constitutive element. Hinton and Nowlan [6] demonstrate that learning can guide evolution (the Baldwin effect) without formalizing the threshold that governs promotion. Finn et

al. [4] and Thrun [18] develop meta-learning algorithms that operate as M on M, the $M^{(2)}$ level in our hierarchy, but do not derive why the meta-level necessarily learns more slowly.

2 Formal Framework

We introduce the minimal definitions required for the two main results, extending the framework of [10].

Definition 1 (Entity, World, Uncertainty)**.** An entity $E = (K, A, \sigma, \gamma)$ consists of a knowledge base K (propositions about world W), an action space A (executable actions), a selection mechanism σ that chooses actions given context, and an agency parameter γ. The world state space W is a measurable space; the true state $w \in W$ is not directly observable. The *uncertainty* of E about state w given knowledge K is:

$$U(w, K) = -\log P(w \mid K)$$

The expected uncertainty is the conditional entropy:

$$H(W \mid K) = \mathbb{E}_w[U(w, K)] = -\sum_w P(w \mid K) \log P(w \mid K)$$

Definition 2 (Agency)**.** The agency parameter $\gamma \in [0, 1]$ is the probability that E engages its learning mechanism M in response to a learning signal. At $\gamma = 0$, A and K are fixed forever: the entity receives signals but never updates. The operational signature of $\gamma > 0$ is that the entity bounds uncertainty beyond what immediate survival requires, freeing active context capacity for novel states. An entity that updates only under immediate threat has $\gamma \approx 0$; an entity that builds reserve capacity for states it has not yet encountered has $\gamma > 0$.

Definition 3 (Action Feedback)**.** For action a executed in world state w, the *feedback* $\varphi(a, w)$ is the observable consequence. The entity updates its knowledge base: $K_a = K \cup \{p_a\}$, where p_a is the proposition encoding the feedback. The *gradient signal* is the prediction error:

$$\delta = \varphi(a, w) - \mathbb{E}[\varphi(a, \cdot) \mid K]$$

Definition 4 (Information Gain)**.** The information gain from executing a given K is the mutual information between the action's outcome and the world state, conditioned on current knowledge:

$$G(a, K) = I(a; W \mid K) \geq 0$$

Non-negativity follows from the definition of mutual information [2]. $G = 0$ if and only if a provides no information about W beyond what K already contains: the encounter is not novel.

Definition 5 (Novelty Rate)**.** The *novelty rate* $\nu(E)$ is the rate at which entity E encounters world states w where $U(w, K) > 0$. ν is a property of the entity-environment coupling, not of W alone: the same environment presents different novelty rates to entities with different K. $\nu(E) > 0$ whenever the entity's environment is not fully predicted by K.

Definition 6 (Encoding Hierarchy)**.** The encoding space $\mathcal{E}(c, r)$ is a continuous space parameterized by cost c and reversibility r. Within this space, interfaces $L_0, L_1, \ldots, L_n$ define discrete encoding levels with increasing cost and reversibility:

- L_0 (genomic): highest cost of error, lowest reversibility, slowest update rate.
- L_n (active context): lowest cost of error, highest reversibility, fastest update rate.

Escalation [11] pushes proven actions upward toward more permanent levels. The present paper derives the threshold that governs this promotion.

3 Gradient Descent on Uncertainty

Theorem 1 (Evaluation-Driven State Revision)**.** *Let $E = (K, A, \sigma, \gamma)$ be an entity with $\gamma > 0$ persisting in world W under bounded active context C_n.*

Part I. *Every executed action generates a gradient signal, and $\gamma > 0$ guarantees engagement with that signal, producing evaluation-driven state revision, gradient descent on U, with convergence to a local minimum.*

Part II. *Under positive novelty rate $\nu(E) > 0$ (Definition 5), the gradient is never globally zero: the entity continues to encounter states where $U(w, K) > 0$, so descent never terminates.*

Proof. **Part I: Forced Signal, Response Gate, and Convergence.**

Every executed action a generates feedback $\varphi(a, w)$ from the world. This signal arrives whether or not the entity has $\gamma > 0$: the world responds to actions regardless of the actor's capacity to learn from them. The gradient signal $\delta = \varphi(a, w) - \mathbb{E}[\varphi(a, \cdot) \mid K]$ is therefore produced at every action.

Agency γ is the response gate. An entity with $\gamma = 0$ receives every signal δ but never updates: (A, K) remain static. An entity with $\gamma > 0$ engages M with probability γ on each signal, producing the update:

$$\Delta_a = -\eta \cdot \delta$$

where $\eta > 0$ is the learning rate of mechanism M. The update $\Delta_a = -\eta \cdot \delta$ revises K in the direction that reduces prediction error. K is a set of propositions, not a parameter vector, so $\nabla_K U$ is not defined in the classical sense. The descent property is instead established directly: Step 1 below shows that each informative update strictly reduces $\mathbb{E}[U]$, using the fact that conditioning on additional evidence cannot increase expected uncertainty [2]. This is evaluation-driven state revision: the entity updates in a descent direction on U without requiring a differentiable parameterization of K.

We now establish convergence via a scalar Lyapunov argument that requires no dimensional decomposition.

Step 1: Each M-update from a novel encounter strictly reduces $\mathbb{E}[U]$. When the entity encounters a state w where $U(w, K) > 0$ and executes action a, the encounter is informative: $G(a, K) > 0$ because the feedback resolves uncertainty that K did not already contain. When M integrates this feedback into K, forming $K' = K \cup \{p_a\}$:

$$\mathbb{E}[U(w, K') \mid K] \leq U(w, K) - G(a, K)$$

The inequality is strict whenever $G(a, K) > 0$—that is, whenever the encounter was novel. Novelty drives the descent: each novel encounter produces an informative update, and each informative update strictly reduces expected uncertainty.

Step 2: Monotone convergence. Consider the sequence $\{E_t\} = \{\mathbb{E}[U(w, K(t))]\}$ indexed by update steps t. By Step 1, $\{E_t\}$ is non-increasing. By definition, $U(w, K) = -\log P(w \mid K) \geq 0$, so $\{E_t\}$ is bounded below by 0. By the monotone convergence theorem, $\{E_t\}$ converges. Since the sequence is non-increasing and convergent:

$$\sum_{t=0}^{\infty} G(a_t, K(t)) \leq U(w, K(0)) < \infty$$

Therefore $G(a_t, K(t)) \to 0$ as $t \to \infty$. The entity converges to a local minimum of U: a state where available actions yield diminishing informational returns.

Step 3: Dimensionality irrelevance. The entire argument operates on the scalar quantity U. The entity's internal parameterization may grow as the reachable trajectory space $T_{\text{reachable}}$ expands. Each new proposition p_a added to K from an informative encounter can only reduce $\mathbb{E}[U]$ (Step 1): integrating novel evidence never increases expected uncertainty. Cross-grounding between propositions accelerates convergence by allowing a single observation to update multiple related beliefs. Adding structure to K is analogous to adding a column to a design matrix and re-normalizing: the residual can only shrink. The Lyapunov bound $U \geq 0$ holds regardless of the dimensionality of K or the parameterization of M.

Part II: Non-Termination Under Positive Novelty.

When $\nu(E) > 0$ (Definition 5), the entity encounters states w with $U(w, K(t)) > 0$ at positive rate. Each such encounter produces an action a with $G(a, K(t)) > 0$: the feedback from acting under uncertainty is informative. Therefore at every finite t there exists some a' with $G(a', K(t)) > 0$.

The gradient is never globally zero. Descent converges locally (Part I) but never terminates globally: as long as the environment presents novel states, there remain actions with positive information gain.

Non-termination does not require unbounded world complexity. A finite world with $\nu(E) > 0$ suffices: even if W is finite, the entity may never fully predict it because new evidence can invalidate or refine existing propositions, and the combinatorial space of entity-environment interactions exceeds what K can index under bounded C_n.

Interaction with the curation regime. The convergence argument in Part I assumes K grows monotonically ($K' = K \cup \{p_a\}$). In the curation regime established in [12], $|K|$ is bounded at $K_{\max}$ and every addition requires displacement. Does displacement break convergence?

Not under the cost-benefit displacement criterion of the K/A inseparability theorem. Every proposition $p \in K$ incurs a positive maintenance cost $c_{\text{retain}}(p, K) > 0$: retrieval interference, edge maintenance, and context consumption. Proposition p is displaced by q only when

$$\mathbb{E}[B(q, \mathcal{T})] - c_{\text{retain}}(q, K) \;>\; \mathbb{E}[B(p, \mathcal{T})] - c_{\text{retain}}(p, K),$$

where B is expected survival benefit and $\mathcal{T}$ is the entity's task distribution. This is a *net* criterion: it accounts for the grounding lost when p is removed, not just the grounding gained by adding q. Displacement occurs only when the swap strictly improves net survival value. Therefore $\mathbb{E}[U]$ does not increase across displacements, and convergence is preserved—not because K grows, but because each displacement strictly improves the net survival benefit of a fixed-size K.

Remark. An entity with $\gamma = 0$ never updates. As W changes, its fixed K becomes increasingly inaccurate: $U(w_t, K)$ grows until it exceeds the lethality threshold U_{lethal}. Selection removes such entities. Therefore $\gamma > 0$ is necessary for persistence, a claim established fully in Paper 5. □

4 Multi-Timescale Structure

Corollary 2 (Individual Learning and Evolution Share Update Structure)**.** *Both individual learning and evolutionary selection instantiate the same abstract update: execute, receive signal, retain or revise. They differ in mechanism, object, and timescale.*

Proof. **Individual learning.** The entity updates its knowledge and action space directly from continuous prediction-error signals δ. Each update revises K in a direction that reduces $\mathbb{E}[U]$

(Theorem 1, Part I). The update is directed: δ specifies both magnitude and direction of revision. The learning rate η_{ind} is determined by the capacity of mechanism M.

Evolutionary selection. The population-level update operates on a different object (the heritable component of A) via a different mechanism:

$$\Delta a_{\text{species}} = \text{select}(\text{mutate}(A_{\text{population}}))$$

Mutation is undirected: random perturbation of heritable parameters. Selection is binary: survive or not. No individual organism descends a gradient at the evolutionary timescale; the population distribution shifts toward lower U through differential survival.

Both processes move the entity (individual or lineage) toward lower U at their respective timescales. Gradient descent is the efficient limit: directed δ combined with capable M produces the steepest feasible descent. Evolution is the general case: random perturbation combined with selection produces descent without requiring any individual to compute a gradient.

The two are not independent. Individual learning discovers actions at L_n that, if sufficiently validated, promote to L_0 via the encoding hierarchy (Theorem 3). This is the Baldwin effect [6], here derived rather than observed. □

5 Encoding Permanence

Theorem 3 (Encoding Permanence Scales with Rigor)**.** *Let $L_0, \ldots, L_n$ be encoding levels with associated failure costs $C_0 > C_1 > \cdots > C_n$ and reversibility costs $r_0 > r_1 > \cdots > r_n$. Let ε be the maximum acceptable expected error cost. Then the minimum certainty threshold for encoding an action at level L_i is:*

$$\theta_i = 1 - \frac{\varepsilon}{C_i}$$

Proof. Encoding action a at level L_i has reversibility cost r_i: correcting a bad encoding is not impossible but expensive in proportion to r_i (at L_0, correction requires generations of selection; at L_n, a single update suffices). If the encoding is erroneous—a is not in fact reliably adaptive—the cost of that error is C_i. If θ is the certainty that a is correct, then the expected error cost is:

$$\mathbb{E}[\text{error cost}] = (1 - \theta) \cdot C_i$$

Encoding is safe when the expected error cost does not exceed the acceptable threshold:

$$(1 - \theta) \cdot C_i \leq \varepsilon$$

Rearranging:

$$\theta \geq 1 - \frac{\varepsilon}{C_i}$$

The minimum safe threshold is therefore $\theta_i = 1 - \varepsilon/C_i$. This is the break-even point on a cost continuum: the certainty at which the expected benefit of encoding at L_i exceeds the expected cost of error weighted by C_i.

Consequence. At L_0 (genomic encoding), C_0 is the cost of lineage extinction, orders of magnitude larger than ε. Therefore $\theta_0 \approx 1$: near-certainty is required before the expected benefit justifies the risk. A bad encoding at L_0 does not kill one individual; it kills the lineage. At L_n (active context), C_n is small (cost of a single failed action), so θ_n is meaningfully less than 1: the expected benefit of trying exceeds the expected cost of failure at moderate confidence. Rigor is proportional to reversibility cost. □

Error propagation through the hierarchy. A natural objection holds that errors compound across levels: if each level has error rate $1 - \theta_i$, the cumulative error grows multiplicatively. This objection is based on a naive cascading model. Under escalation [11], errors partition into two categories: *handled errors*, which are resolved at level L_i and never reach L_{i+1}, and *unhandled exceptions*, which represent genuinely irreducible failures that escalate. The error rate seen by L_{i+1} is not a compounded Bernoulli product of lower-level error rates but the rate of genuinely irreducible errors: failures that persist across sufficient variation to pass the threshold θ_i. The thresholds θ_i are independent by construction: each is determined by C_i and ε, not by the error rates of adjacent levels.

5.1 Deriving the Learning Rate Hierarchy

The learning rate at level L_i is inversely related to the rigor threshold. To see why, consider the sample complexity: encoding at L_i requires confidence θ_i that the action is adaptive. By Hoeffding's inequality [7], the number of independent trials n_i needed to achieve confidence θ_i satisfies:

$$n_i \;\geq\; \frac{\log(1/(1-\theta_i))}{2\varepsilon^2} \;\approx\; \frac{1}{2\varepsilon^2}\log\frac{C_i}{\varepsilon}$$

The effective learning rate, the fraction of experience that results in permanent encoding, is inversely proportional to n_i:

$$\eta_i \;\propto\; \frac{1}{n_i} \;\propto\; \frac{1}{\log(C_i/\varepsilon)}$$

For L_n (active context): C_n is small but finite, $\theta_n = 1 - \varepsilon/C_n$ is meaningfully less than 1, and η_{ind} is large. The entity updates rapidly from each gradient signal.

For L_0 (genomic): $C_0 \gg \varepsilon$, so $\theta_0 \approx 1$ and $\eta_{\text{evo}} \approx \varepsilon/C_0 \approx 0$. The lineage updates only when certainty is overwhelming.

Since $\theta_0 \gg \theta_n$, it follows that $\eta_{\text{evo}} \ll \eta_{\text{ind}}$. The learning rate difference between evolution and individual learning is *derived* from the cost-of-failure structure, not assumed as a design parameter.

Corollary 4 (Bounded Adaptation)**.** *An action a is* certain *at level L_i when $U(w, K_a) \approx 0$ across sufficient variation to meet threshold θ_i. Once θ_i is met, the expected benefit of encoding at L_i exceeds the expected cost of error. The action is encoded at L_i and no longer consumes L_n resources, freeing active context for the next epistemic frontier. This encoding is durable, not literally irreversible: environmental shift can render it maladaptive, but correction requires effort proportional to r_i.*

Definition 7 (Certainty)**.** Action a is *certain* when $U(w, K_a) \approx 0$ across sufficient variation in w to meet the threshold θ_i for encoding level L_i. Formally: $P(a \text{ is adaptive} \mid K) \geq \theta_i$, where the probability is taken over the distribution of world states relevant to L_i.

6 Engagement with Prior Work

Dupoux, LeCun, and Malik [3]. Their bilevel optimization framework, inner loop as development, outer loop as evolution, captures the correct structural relationship between encoding levels. However, they treat the learning rate difference between levels as a design parameter: the inner loop is fast because it is designed to be fast, and the outer loop is slow because mutation and selection are slow. Theorem 3 derives this relationship from first principles: $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ because $C_0 \gg C_n$, not because of any design choice.

Their framework includes a "System M" that is "not learned through gradient descent" but fixed by evolution. Our framework explains *why*: the rigor threshold $\theta_0 \approx 1$ means that any component encoded at L_0 must be near-certain across all environments in the species distribution. Only mechanisms that generalize across the full environmental distribution meet this threshold. System M is not excluded from gradient descent by fiat; it is excluded because the cost of a bad learning mechanism at L_0 is lineage extinction, and the resulting threshold θ_0 permits only components validated across maximal variation.

Hutter [8]: AIXI. AIXI constructs the theoretically optimal agent by enumerating all computable hypotheses, weighted by Kolmogorov complexity. The construction assumes unbounded computation, precisely the constraint our framework takes as central. Under bounded active context C_n, the AIXI solution is unreachable: the entity cannot enumerate hypotheses but must descend the gradient with finite resources. Our Theorem 1 characterizes what bounded agents actually do: local gradient descent on U with convergence to local minima, not global optimization over hypothesis space.

Chollet [1]: Intelligence as skill-acquisition efficiency. Chollet's measure of intelligence, the rate at which an agent acquires new skills from limited experience, is structurally close to our learning rate η_M. The key divergence is in the treatment of prior knowledge. Chollet treats priors as a confound to be controlled for: intelligence is efficiency *net of* prior knowledge. In our framework, prior knowledge K is not a confound but a constitutive element: η_M operates on K, and the capacity of M is inseparable from the structure of K [12]. An entity with richer K does not have "unfair advantage." It has done more gradient descent.

Schmidhuber [15]: Curiosity as compression progress. Schmidhuber defines intrinsic motivation as the rate of compression progress: the reduction in description length of the agent's model of the world. This is structurally identical to η_M: both measure the rate of uncertainty reduction. The difference is ontological. Schmidhuber treats curiosity as a design objective, something to be engineered into agents. Our framework suggests that $\gamma > 0$ (the tendency to engage M even without immediate survival pressure) is not a design choice but a necessary condition for persistence, a claim established fully in Paper 5.

Hinton and Nowlan [6]: Learning guiding evolution. Hinton and Nowlan demonstrate via simulation that individual learning can accelerate evolutionary search: the Baldwin effect. An organism that can learn a trait during its lifetime effectively smooths the fitness landscape for evolution. Our framework formalizes this: individual learning at L_n discovers actions via rapid gradient descent (η_{ind} large). Encoding permanence (Theorem 3) determines which of these actions promote to L_0: only those meeting $\theta_0 \approx 1$, actions validated across sufficient environmental variation to warrant heritable encoding. The Baldwin effect is not an empirical curiosity but a necessary consequence of the encoding hierarchy.

Kirkpatrick et al. [9]: Catastrophic forgetting and elastic weight consolidation. Kirkpatrick et al. demonstrate that neural networks trained sequentially on multiple tasks catastrophically forget earlier tasks unless protected by a consolidation mechanism. Their solution, elastic weight consolidation (EWC), assigns higher "importance" to weights encoding well-established knowledge, penalizing changes to those weights during subsequent learning. This is precisely the encoding hierarchy: weights at lower encoding levels (higher θ_i, higher cost of error) are protected; weights at higher levels (lower θ_i) are free to change. EWC's Fisher information matrix empirically estimates our cost-of-failure C_i. The result provides direct empirical evidence that the encoding hierarchy is not merely theoretical: artificial systems that violate it (flat learning rate across all parameters) fail, and systems that approximate it (EWC, progressive neural networks) succeed.

Murray et al. [13]: Cortical temporal hierarchy. Murray et al. establish that intrinsic timescales of neural activity increase systematically along the cortical hierarchy: sensory cortex fluctuates rapidly, while prefrontal cortex maintains stable representations over seconds. This gradient of temporal stability maps directly onto the encoding hierarchy: fast-updating sensory areas operate at L_n (low cost of error, high reversibility), while slow-updating prefrontal areas operate at lower encoding levels (high cost of error, low reversibility). The empirical timescale gradient matches the theoretical prediction $\eta_i \propto 1/\log(C_i/\varepsilon)$: regions with higher cost of error update more slowly.

Finn et al. [4] and Thrun [18]: Meta-learning. Model-Agnostic Meta-Learning (MAML) learns an initialization from which task-specific learning converges rapidly. "Learning to learn" [18] frames this as optimizing the learning algorithm itself. In our framework, meta-learning is M operating on M: the $M^{(2)}$ level. The meta-learner sits at a higher encoding level than the base learner: it updates more slowly ($\eta_{M^{(2)}} < \eta_M$) because the cost of a bad meta-learning rule ($C_{M^{(2)}}$) exceeds the cost of a bad base-level update (C_M). The MAML inner loop is L_n; the outer loop is L_{n-1}. The hierarchy extends: evolution is meta-meta-...-learning at L_0.

7 Conclusion

Gradient descent on uncertainty is not a metaphor for learning. It is what survival-driven updating *is*. Any entity with $\gamma > 0$ that persists under bounded context necessarily engages in evaluation-driven state revision: executing actions, receiving gradient signals, and updating knowledge to reduce U. The convergence argument is scalar; it operates on U directly, with no dependence on the dimensionality of the entity's internal parameterization. Under positive novelty rate ($\nu(E) > 0$), the gradient never terminates: novel states ensure actions with positive information gain always exist.

The encoding hierarchy is not a design choice but a necessary consequence of cost-of-failure differences across encoding levels. The threshold $\theta_i = 1 - \varepsilon/C_i$ is the break-even point: the certainty at which expected benefit of encoding exceeds expected cost of error. From this single equation, the entire learning rate hierarchy follows: $\eta_{\text{evo}} \ll \eta_{\text{ind}}$ because $C_0 \gg C_n$, not because evolution is "inherently slow" or individual learning is "inherently fast."

Together with graph necessity [10], convergence to escalation [11], and K/A inseparability [12], this paper completes the structural account of how knowledge is acquired, validated, and encoded. What remains is to establish that $\gamma > 0$ is itself necessary for persistence: that agency is not optional but required. This is the subject of Paper 5.

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